

- 12 A sample of 1 mol of XY was placed in an empty 1 dm³ container and allowed to reach equilibrium with a total pressure, p , according to the following equation.



At equilibrium, x mol of XY had dissociated. What is the value of the equilibrium constant, K_p , at the temperature of the experiment?

- A $\frac{x^2}{(1-x)^2}$
- B $\frac{(1-x)^2}{x^2}$
- C $\frac{4(1-x)^2}{x^2}$
- D $\frac{x^2}{4(1-x)^2}$